Spring potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate stored spring energy and distinguish it from force.

SUCCESS CRITERIA - I can...

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “compression” and “spring constant” to explain your answer. Sentence starters are optional.

compression: shortening or squeezing relative to an unstressed length

spring constant: how much an ideal spring’s force changes per stretch

potential energy: energy associated with the configuration of interacting objects

01 Warm-up: bring back an earlier idea

A cart starts from rest at a height of **3 m**. Can it reach **5 m** with no energy input from outside and no losses? Use kinetic energy to **explain your answer**.

Word help: kinetic energy = energy associated with motion

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

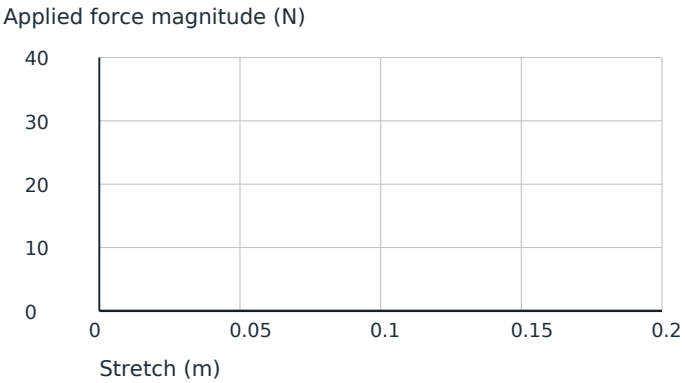
02 Worked example: explain the prediction

Reference: $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

Worked reasoning: A spring with k=**200 N/m** stretched **0.10 m** stores $\frac{1}{2}kx^2$=**1 J**. Doubling stretch to **0.20 m** stores **4 J**, not **2 J**.

Your turn: Draw force-versus-stretch triangles and **compare their areas**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The same spring is displaced x and **2** x. **Compare stored energies** and force magnitudes.

Sentence starter: Both ... . They differ in ... because ... .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Apply the physics

A spring launcher has k = **200 N/m** and compression **0.10 m**. It launches a **0.50 kg** cart along a frictionless horizontal track.
1. **Label the spring diagram**.
2. **Calculate stored energy** and the cart’s speed when the spring reaches its natural length.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is energy positive for both positive and negative x even though spring force reverses?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **8 N** force acts at **120°** to a **6 m** displacement.
Calculate its work and **explain the sign**.
Label force and displacement, including their relative direction.

Word help: work = energy transferred by a force along a displacement

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

k=**80 N/m**; stored energy **0.40 J**. **Find displacement magnitude** and force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.5 kg** cart is launched by a spring k=**400 N/m** compressed **0.15 m** with no losses.
1. **Find release speed**.
2. What compression doubles that speed?

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3